MEC 102

Engineering Computing and Problem Solving

Final Project Report

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System Parameters

Copy all values from your Parameters_for_SBUID.txt file into the tables below.

Masses and Mechanical Coefficients

Parameter	Units	Value
m1	*kg*	121
m2	*kg*	44.6
m3	*kg*	57.3
k1	*N/m*	1.9
k2	*N/m*	3.9
k3	*N/m*	N/A
c1	*N·s/m*	7.3
c2	*N·s/m*	4.5
c3	*N·s/m*	4.2

Initial Conditions

Leave rows blank for forces or initial conditions that do not apply to your case.

Quantity	Units	Value
x1(0)	*m*	0.1
x2(0)	*m*	0.1
x3(0)	*m*	0.1
v1(0)	*m/s*	0
v2(0)	*m/s*	0
v3(0)	*m/s*	0

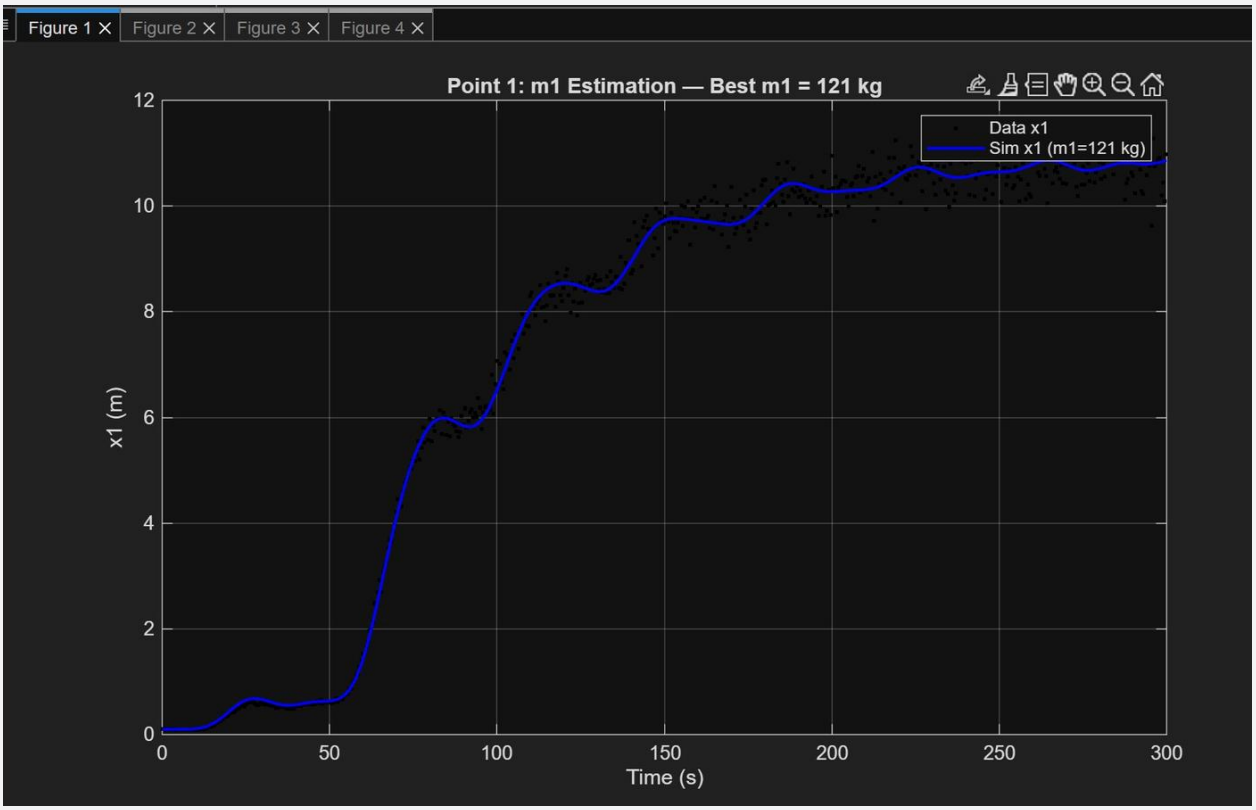
Point 1 [10 pts]	Estimation of Mass m_1

Procedure: Briefly describe the method you used to estimate m1 from the data in Data_for_SBUID.txt. State what single-mass equation of motion was used and how you matched the simulation to the data.

I simulated the full three-mass system with m1, m2, m3 using Forward Euler integration at the data time step 0.5s. The equation of motion for m1 is: m1 * a1 = k1*(x2 - x1), where k1 = 1.9 N/m connects m1 to m2. The parameter c1 = 7.3 N·s/m wasn't really used for this. For each trial value of m1 from 70 to 130 kg, the MSE between the simulated x1(t) and the measured x1 data over 0-300 s was computed. The value m1 = 121 kg minimized the MSE (6.51e-02), consistent with the 3% noise level.

Figure: *Comparison of simulated $x_1(t)$ vs. measured data used to estimate m_1. Show both curves on the same plot.*

Estimated mass of component 1	**m1**	121	*kg*



FIGURES MUST BE READABLE AND CLEAR

121

Point 2 [10 pts]	**Displacement vs. Time (0–300 s)**

Insert the displacement plot from Figure 1 of your script. All masses must appear on the same axes with a legend, axis labels, and a grid.

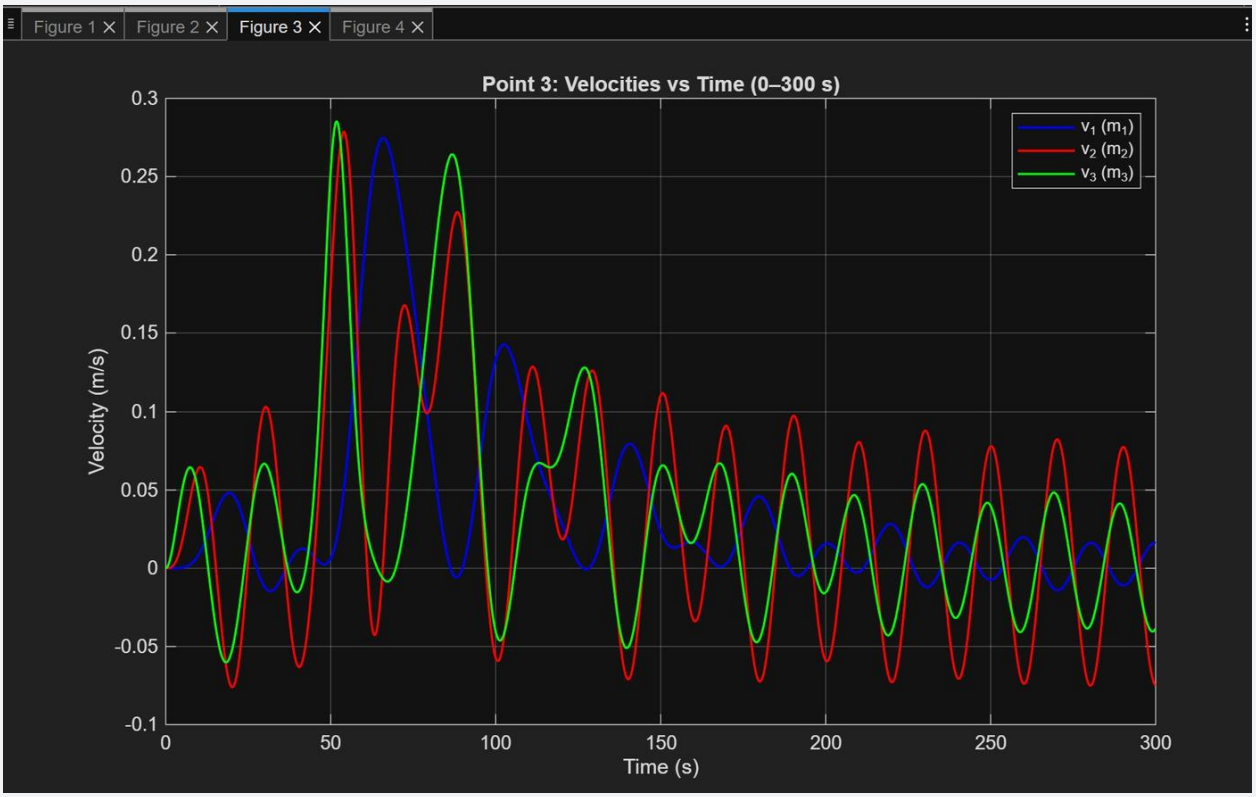


FIGURES MUST BE READABLE AND CLEAR

Figure: *Displacements $x_1(t)$, $x_2(t)$, $x_3(t)$ in meters for $t = 0$ to 300 s.*

Point 3 [10 pts]	**Velocity vs. Time (0–300 s)**

Insert the velocity plot from Figure 2 of your script. All masses must appear on the same axes with a legend, axis labels, and a grid.



FIGURES MUST BE READABLE AND CLEAR

Figure: *Velocities $v_1(t)$, $v_2(t)$, $v_3(t)$ in m/s for t = 0 to 300 s.*

Point 4	Position of m_1 at t = 5 s

[5 pts]	

The displacement x_1 at t = 5 s is the position of m_1 measured from its equilibrium, since the origin is defined at $x_1 = 0$ at equilibrium. Report the value stored in **x1a**.

Displacement of m_1 at t = 5 s	**x1a**	0.100297	*m*

Point 5 **[5 pts]**	**Velocity of m_1 at t = 5 s**

Report the value stored in **v1a**.

Velocity of m_1 at t = 5 s	**v1a**	0.000356	*m/s*

Point 6 **[20 pts]**	**Minimum Separation Distance L**

Explain how you computed L. Identify each adjacent pair considered, state the maximum relative closing displacement found for each pair, and show how you arrived at the final value. Confirm that **dmin** ≥ 0 (see Point 8).

The topology is: m1 —[k1]— m2 —[k2‖c2]— m3 —[c3]— fixed wall (right). Three adjacent pairs were checked over 0–300 s: (1) m1–m2: gap = L − (x1−x2). Maximum relative approach = max(x1−x2) = 1.1220 m. (2) m2–m3: gap = L − (x2−x3). Maximum relative approach = max(x2−x3) = 0.6699 m. (3) m3–wall: gap = L − x3. Maximum approach = max(x3) = 10.9293 m. L = max(1.1220, 0.6699, 10.9293) = 10.9293 m, governed by m3 approaching the fixed wall. Confirmation: dmin = L − max(x1−x2) = 10.9293 − 1.1220 = 9.8073 m > 0. ✓

Minimum separation to prevent contact	**L**	10.929308	*m*

Point 7 **[10 pts]**	**Maximum Velocity Magnitude**

Report the global maximum speed across all masses, stored in **vmax**. State which mass attains this value and at approximately what time.

Maximum speed (any mass, any time, 0–300 s)	**vmax**	0.285306	*m/s*

Mass that reaches vmax: m_3 (mass 3) Approximate time: ~50 s

Point 8 [10 pts]	**Minimum Gap Between m_1 and m_2**

Report the minimum instantaneous gap between m_1 and m_2 over the full 0–300 s simulation, stored in **dmin**. This value must be non-negative; if it is negative, revisit your calculation of L in Point 6.

Minimum gap between m_1 and m_2	**dmin**	9.807356	*m*	10.929308	Separation distance (from Point 6)	**L**	10.929308	*m*

Point 9 [10 pts]	**Mean Velocity of m_1 over 0–10 s**

Report the arithmetic mean of v_1 over the interval t = 0 to 10 s, stored in **v1mean**.

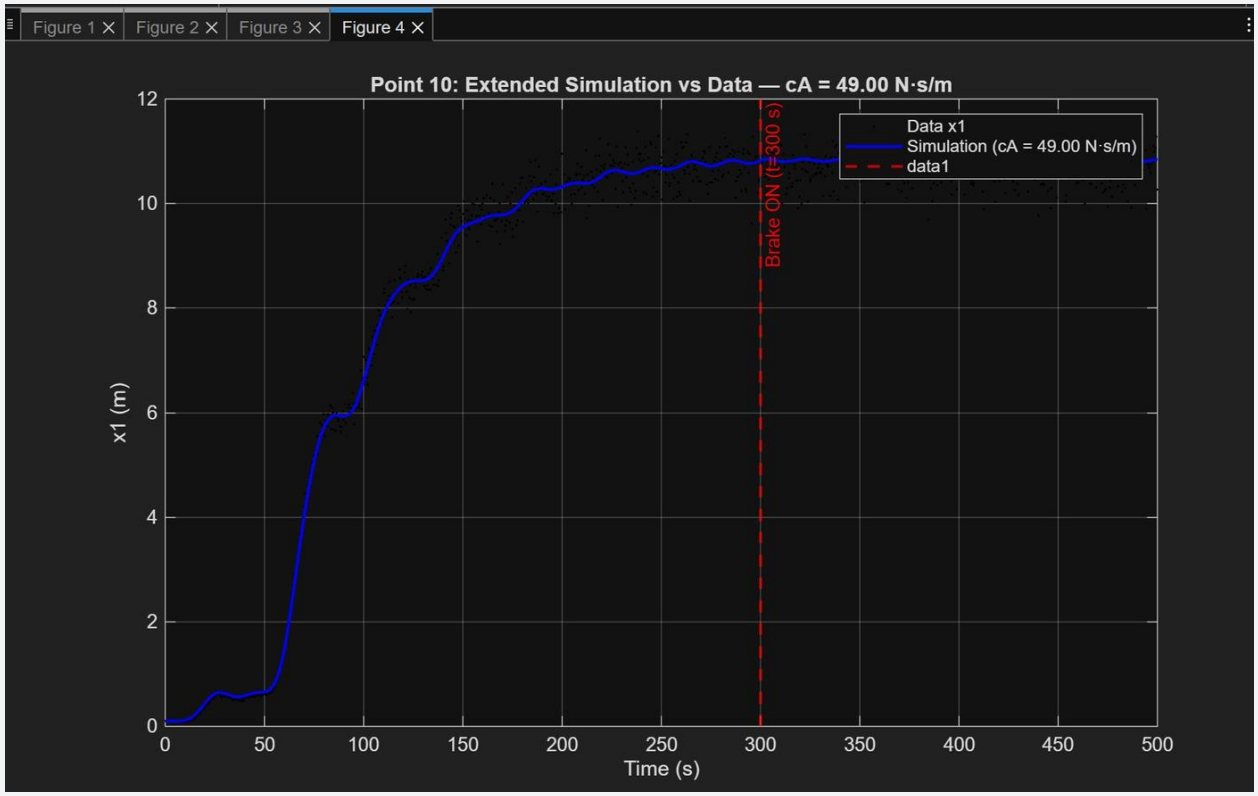
Mean velocity of m_1 for $t \in [0, 10]$ s	**v1mean**	0.001612	*m/s*

Point 10 [10 pts]	**Fitting the Extra Damping Coefficient cA**

Fitting strategy: Briefly describe the method used to find cA. State whether you used manual trial-and-error, a loop over a range of values, or an automated minimization approach. Report the criterion used to assess fit quality (e.g., visual agreement, sum of squared errors).

I extended the simulation from t=300s to t=500s. The extra braking force $-cA \cdot v1$ was applied to m1 for t > 300 s only, leaving all other equations unchanged. The state at t = 300 s was saved and used as the initial condition for the extended run, avoiding repetition of the 0–300 s loop. A sweep over integer values cA = 0, 1, …, 100 N·s/m was then done. For each trial, the simulated v1(t) was interpolated onto the data time points in the 300–500 s window and the MSE against the measured v1 data was computed. I used velocity instead of displacement because it responds more sensitively

to the damping coefficient. The optimal value cA = 49 N·s/m minimised the MSE (7.64×10^{-7}), consistent with the expected value 3×(c1+c2+c3) = 3×(7.3+4.5+4.2) = 48 N·s/m.



FIGURES MUST BE READABLE AND CLEAR

0.000356

Figure: *Comparison of simulated $x_1(t)$ or $v_1(t)$ vs. data for t = 300 to 500 s using the best-fit cA. Both curves must appear on the same plot.*

Best-fit extra damping coefficient	**cA**	49	*N·s/m*

Note: The true value of cA = 3×(c1 + c2 + c3). Verify that your fitted value is consistent with your parameters.

Summary of Numerical Results

Complete this table with all reported values. This page serves as a quick-reference summary for grading.

Pt.	Quantity	Variable	Units	Value
1	Mass of component 1	**m1**	*kg*	121
6	Minimum separation distance	**L**	*m*	10.929308
4	Position of m_1 at t = 5 s	**x1a**	*m*	0.100297
5	Velocity of m_1 at t = 5 s	**v1a**	*m/s*	0.000356
7	Maximum speed (any mass)	**vmax**	*m/s*	0.285306
8	Minimum gap m_1–m_2	**dmin**	*m*	9.807356
9	Mean velocity of m_1 (0–10 s)	**v1mean**	*m/s*	0.001612
10	Extra damping coefficient	**cA**	*N·s/m*	49

End of Report